

A nilpotent right-perturbation criterion for hypercyclic generalized derivations

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2026-06-21

Status

This is a likely valid full positive answer to Section 6, Question 1 of arXiv:1605.07409, Clifford Gilmore, *Dynamics of Generalised Derivations and Elementary Operators*. It gives a broad positive answer to the first question in Section 6 of that paper, which asks whether reasonable sufficient conditions exist on the pair (A, B) . It is not a characterization of all hypercyclic generalized derivations.

Source question

The source paper defines, for operators $A, B \in \mathcal{L}(X)$,

$$\tau_{A,B}(S) = L_A(S) - R_B(S) = AS - SB.$$

In Section 6, Question 1 asks:

Do reasonable sufficient conditions exist on the pair (A, B) that induce hypercyclic $\tau_{A,B}$ on separable Banach ideals?

The following crop shows the question in the source PDF.

6. FURTHER QUESTIONS

Some natural questions arising from this paper include the following.

1. Do reasonable sufficient conditions exist on the pair (A, B) that induce hypercyclic $\tau_{A,B}$ on separable Banach ideals?
2. Corollary 5.2 and Example 5.3 contrast the hypercyclic properties of

Idea of the proof

The source paper already shows how to turn one-sided hypercyclicity into hypercyclic generalized derivations when the second operator is a scalar multiple of the identity. The point here is that scalarity can be replaced by nilpotence. Suppose A has a right inverse F with spectral radius less than one, and suppose a dense subspace is eventually killed by powers of A . If $B = N$ is nilpotent, then R_N is a commuting nilpotent perturbation of the left multiplier L_A . The expression

$(L_A - R_N)^n L_F^n$ is an invertible polynomial in the nilpotent operator $L_F R_N$. Its inverse grows at most polynomially in n , while L_F^n decays exponentially. This gives explicit approximate inverse maps for the Hypercyclicity Criterion. On finite-rank operators whose ranges lie in the eventually killed dense subspace, the powers of $L_A - R_N$ are eventually zero.

Theorem

Let X be a Banach space, let $J \subset \mathcal{L}(X)$ be a separable admissible Banach ideal in the sense used in the source paper, and let $A, N \in \mathcal{L}(X)$. Assume that:

1. $N \neq 0$ is nilpotent;
2. there is $F \in \mathcal{L}(X)$ such that $AF = I_X$ and $r(F) < 1$;
3. there is a dense linear subspace $X_0 \subset X$ such that, for every $x \in X_0$, $A^n x = 0$ for all sufficiently large n .

Then

$$\tau_{A,N} = L_A - R_N : J \rightarrow J$$

satisfies the Hypercyclicity Criterion. In particular, it is hypercyclic.

Thus every pair (A, N) satisfying the three conditions above is a concrete sufficient condition for the first final question of the source paper, with N nonzero and non-scalar. Since that question asks whether such reasonable sufficient conditions exist, this is a positive answer to the question as stated.

Proof

Let $m \geq 2$ be such that $N^m = 0$. Then $R_N^m = 0$, and all left multipliers commute with all right multipliers. Since $r(F) < 1$, there are constants $C > 0$ and $0 < \rho < 1$ such that

$$\|F^n\| \leq C\rho^n \quad (n \geq 0).$$

Set

$$T = L_A - R_N \quad \text{on } J, \quad U = L_F R_N.$$

For every $n \geq m$, the binomial formula and $R_N^m = 0$ give

$$T^n L_F^n = \sum_{j=0}^{m-1} (-1)^j \binom{n}{j} L_{A^{n-j} F^n} R_N^j = \sum_{j=0}^{m-1} (-1)^j \binom{n}{j} U^j =: Q_n(U).$$

Here we used $A^{n-j} F^n = F^j$, which follows from $AF = I_X$. Since $U^m = 0$ and $Q_n(0) = 1$, the operator $Q_n(U)$ is invertible, with inverse

$$Q_n(U)^{-1} = \sum_{r=0}^{m-1} (I - Q_n(U))^r.$$

The norm of $Q_n(U)^{-1}$ grows at most polynomially in n , because it is a polynomial of degree $m-1$ in the nilpotent U , with coefficients polynomially bounded in the binomial coefficients $\binom{n}{j}$.

Define

$$S_n = L_F^n Q_n(U)^{-1} \quad (n \geq m).$$

Then

$$T^n S_n = T^n L_F^n Q_n(U)^{-1} = I_J.$$

Moreover,

$$\|S_n\| \leq \|L_F^n\| \|Q_n(U)^{-1}\| \leq C \rho^n \|Q_n(U)^{-1}\| \rightarrow 0,$$

since exponential decay dominates the polynomial growth. Hence

$$S_n Y \rightarrow 0 \quad \text{and} \quad T^n S_n Y = Y$$

for every $Y \in J$.

It remains to find a dense subspace on which $T^n \rightarrow 0$. Let

$$\mathcal{D} = \text{span}\{x^* \otimes x : x^* \in X^*, x \in X_0\}.$$

The admissibility assumptions used in the source paper imply that finite-rank operators are dense in J ; since X_0 is dense in X , $\mathcal{D}$ is dense in J , by the same rank-one approximation argument used in the source paper. If $Y \in \mathcal{D}$, then the range of Y is contained in a finite-dimensional subspace spanned by finitely many vectors of X_0 . For each fixed $0 \leq j < m$, the operator $R_N^j Y = Y N^j$ has range contained in the same finite-dimensional subspace. By the eventual-killing hypothesis on X_0 , there is M_Y such that

$$L_{A^{n-j}} R_N^j Y = 0$$

whenever $n - j \geq M_Y$. Using the binomial expansion

$$T^n Y = \sum_{j=0}^{m-1} (-1)^j \binom{n}{j} L_{A^{n-j}} R_N^j Y,$$

we get $T^n Y = 0$ for all sufficiently large n . Thus $T^n Y \rightarrow 0$ for every $Y \in \mathcal{D}$.

The dense subspace $\mathcal{D}$, the sequence $n \mapsto n$, and the maps S_n verify the Hypercyclicity Criterion for $T = \tau_{A,N}$. Therefore $\tau_{A,N}$ is hypercyclic on J .

Concrete example

Let $X = \ell_2$. Let W_λ be λ times the backward shift for some $\lambda > 1$, and let $F = \lambda^{-1}$ times the forward shift. Then $W_\lambda F = I_X$, $r(F) = \lambda^{-1} < 1$, and $W_\lambda^n x = 0$ eventually for every $x \in c_{00}$. Taking

$$N = e_0 \otimes e_1^*,$$

so that $N e_1 = e_0$, $N e_k = 0$ for $k \neq 1$, and $N^2 = 0$, the theorem gives a hypercyclic generalized derivation $\tau_{W_\lambda, N}$ on every separable admissible Banach ideal $J \subset \mathcal{L}(\ell_2)$.

Verification, novelty check, and limitations

The proof uses only the source paper's admissible-ideal framework and the standard Hypercyclicity Criterion. No computational verification is needed.

This packet gives a broad sufficient-condition theorem, not a classification of all hypercyclic pairs (A, B) . It also does not address the second final question in the source paper about contrasting hypercyclic behavior across different ideals.

Before promotion to a full packet, the run’s cheap indexes and local parsed source corpus were searched for “1605.07409”, “generalised derivations”, “generalized derivations”, “elementary operators”, “nilpotent right perturbation”, “ $L_A - R_N$ ”, and related Hypercyclicity Criterion phrases. Targeted web/arXiv searches on 2026-06-21 for the exact source-question phrase and close variants such as “nilpotent right perturbation generalized derivation hypercyclic” found the source paper but no later matching criterion. The novelty confidence is therefore bounded, but no duplicate was found.

References

References

- [1] C. Gilmore, *Dynamics of Generalised Derivations and Elementary Operators*, arXiv:1605.07409.